Module 2 Look Back: Cell Structure and Function

Name _____ Period _____ Date _____

Learning target	Got it	Shaky	Lost
LT 2.1 Explain cell theory and the evidence behind it; tell an observation, hypothesis, theory and law apart.			
LT 2.2 Compare prokaryotic and eukaryotic cells and plant and animal cells; name what each organelle does.			
LT 2.3 Explain how one cell, or cells organized into tissues, organs and systems, keep homeostasis.			
LT 2.4 Draw the phospholipid bilayer and explain why it is selectively permeable.			
LT 2.5 Predict passive and active transport, and which way water moves in hypotonic, hypertonic and isotonic solutions.			

1. (LT 2.1 · SOL Challenge Q72) Statement 1: In every chemical reaction, the total mass of the reactants equals the total mass of the products. Statement 2: All living things are made of cells, the cell is the basic unit of life, and cells come from other cells, an explanation supported by nearly two centuries of evidence. Which describes the two statements correctly?

A Statement 1 is a law, a description of what always happens; statement 2 is a theory, a tested explanation of a broad set of observations
B Statement 1 is a theory and statement 2 is a law, because statement 2 has more evidence
C Both are still hypotheses until someone proves them
D Statement 2 will become a law once scientists collect enough evidence

2. (LT 2.2 · SOL Challenge Q64) A microbiologist examines a single-celled organism from a hot spring. It has a cell membrane, a cell wall, ribosomes and a loop of DNA lying loose in the cytoplasm. It has no nucleus and no mitochondria. How should it be classified?

A As a eukaryote, because it has ribosomes and a membrane
B As a plant cell, because it has a cell wall
C As a prokaryote, because its DNA is not enclosed in a nucleus
D As something other than a cell, because it lacks organelles such as mitochondria

3. (LT 2.3 · SOL Challenge Q69) On a hot, dry afternoon, the tiny pores on the underside of a bean plant's leaves close partway, even though closing them slows photosynthesis. What does this response do for the plant?

A It lets more carbon dioxide into the leaf
B It stops the leaves from making any glucose
C It traps water vapor inside the leaf to cool it
D It reduces water loss, helping the plant keep a stable internal water balance

4. (LT 2.4 · SOL Challenge Q22) Phospholipids in a cell membrane arrange themselves in two layers, with their heads facing outward toward the watery inside and outside of the cell and their tails pointing toward each other in the middle. Why do they line up this way?

A The heads are attracted to water and the tails avoid it, so the tails hide from water between the two layers
B Water pressure pushes the tails into the center
C The tails are heavier than the heads, so they sink to the middle
D The heads carry a negative charge that repels the tails

5. (LT 2.5 · SOL Challenge Q23) A red blood cell is placed in a solution with a higher salt concentration than the cell's own contents. Which word describes the solution, and what happens to the cell?

A Isotonic; the cell stays the same size
B Hypotonic; the cell loses water and shrinks
C Hypertonic; the cell gains water and swells
D Hypertonic; the cell loses water and shrinks

6. (LT 2.5 · SOL Challenge Q71) Glucose is too large and too polar to slip through the lipid part of a red blood cell's membrane. It enters through a carrier protein, moving from the blood, where there is more glucose, into the cell, where there is less. Which describes this movement?

A Active transport, because a protein is involved
B Exocytosis, because glucose enters inside a vesicle
C Passive transport through a protein, with no energy used, because glucose moves from high to low concentration
D Osmosis, because the glucose is dissolved in water

My answers: 1 ____ 2 ____ 3 ____ 4 ____ 5 ____ 6 ____

Fix one. Pick a question you missed or guessed. What idea makes the right answer right?

_____
_____

Before my test or reassessment, the one thing I will review is

_____

Scan to check: tap your answer, read why.